



# **CV180X & CV181X LDC Debugging User Guide**

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**Revision History**

| Revision | Date       | Description                                        |
|----------|------------|----------------------------------------------------|
| 0.1      | 2021/07/14 | Initial                                            |
| 0.1      | 2021/07/18 | Chapter determination, adding table content        |
| 0.1      | 2021/07/20 | Completion of the experimental diagram and content |

# 1 Disclaimer

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# 2 LDC Function and Specification Description

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The Lens Distortion Correction (LDC) system corrects and expands a frame for Barrel Distortion and Pincushion Distortion, correcting the distorted images in both categories.

## 2.1 Algorithm Specification for Each Processor

If necessary, the reference data should be presented in a table when available.

# 3 LDC Debugging Guide

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## 3.1 Basic Concept

### 3.1.1 Field of View

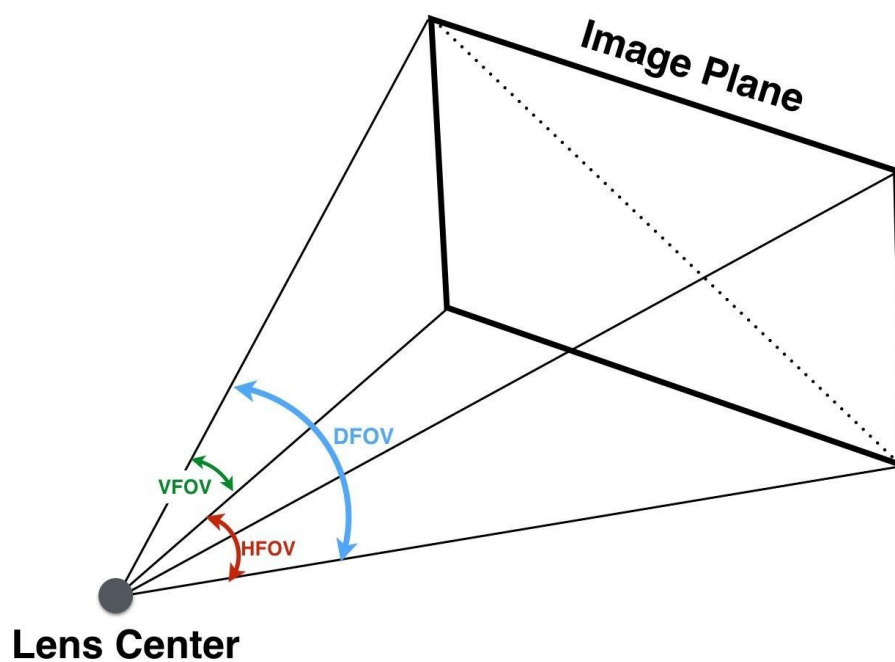


Fig. 3.1: horizontal field of view, vertical field of view, diagonal field of view

## 3.2 Debugging Guide for Each Application Scenario Parameter

### 3.2.1 LDC

Table 3.1: LDC Parameter Configuration Table

| Configuration Parameter | Configuration Range | Parameter Meaning                                                                           |
|-------------------------|---------------------|---------------------------------------------------------------------------------------------|
| CenterXOffset           | -511~+511           | Horizontal offset of the image center point from the physical center point                  |
| CenterYOffset           | -511~+511           | Vertical offset of the image center point from the physical center point                    |
| DistortionRatio         | [-300,500]          | Correction strength, negative numbers for pincushion type, positive numbers for barrel type |
| bAspect                 | bool                | Whether the aspect ratio is maintained during the field of view adjustment                  |
| XYRatio                 | 0~100               | Parameter for field of view size, valid when bAspect=1                                      |
| XRatio                  | 0~100               | X-directional field of view size parameter, valid when bAspect=0                            |
| YRatio                  | 0~100               | Y-directional field of view size parameter, valid when bAspect=0                            |
| stGridInfoAttr          | /                   | GridInfo parameter                                                                          |

Table 3.2: GridInfoAttr's configure parameters

| Configure Parameters | Configuration Range | Parameter Significance      |
|----------------------|---------------------|-----------------------------|
| bEnable              | bool                | Whether to enable GridInfo. |
| gridFileName         | /                   | GridInfo file name.         |
| gridBindName         | /                   | GridInfo binding name.      |
| isBlending           | bool                | Not used at the moment.     |
| bEISEnable           | bool                | Not used at the moment.     |
| homoRgnNum           | /                   | Not used at the moment.     |

### 3.2.2 LDC Correction Model

LDC supports two correction modes, barrel distortion and pincushion distortion, as shown in Fig. 3.2 and Fig. 3.4 .

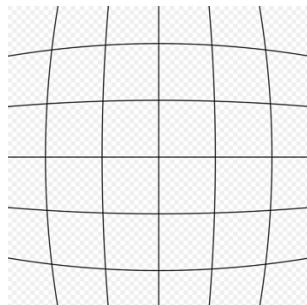


Fig. 3.2: Barrel Distortion

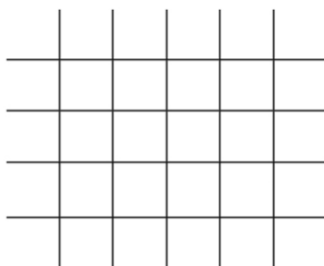


Fig. 3.3: Without Distortion

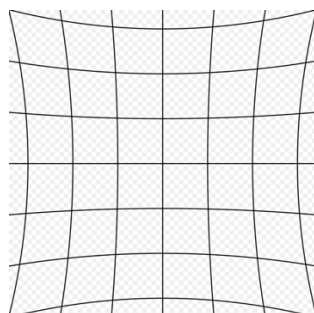


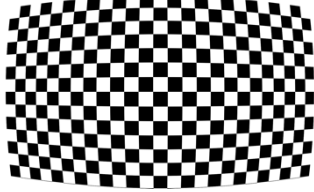
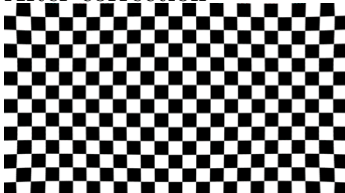
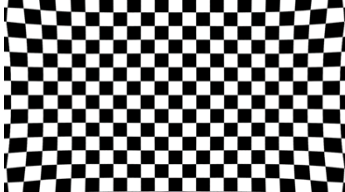
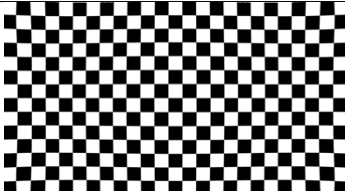
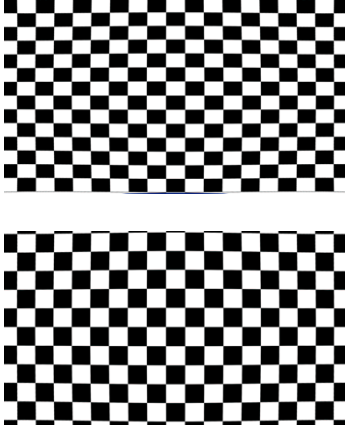
Fig. 3.4: Pincushion Distortion





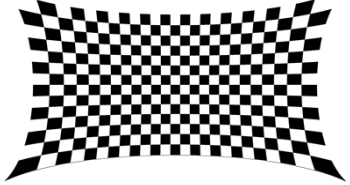
## 3.2.2.1 Example of Correction of Barrel Distortion

Table 3.3: Instructions for correction of barrel distortion

| Parameter Description                                                                                                                                                                                                                                                                                                                                                         | Parameter Configuration                                                                                                                                                       | Image Examples                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Typical Configuration<br>Distortion center overlaps with image center<br>Maintain the aspect ratio<br>Maintain the maximum field of view                                                                                                                                                                                                                                      | Width=1920<br>Height=1080<br>OutWidth=1920<br>OutHeight=1080<br>CenterXOffset/CenterYOffset=0<br>DistortionRatio=-165<br>bAspect=1<br>XYRatio=100<br>XRatio=100<br>YRatio=100 | Before correction<br><br>After correction<br> |
| Ratio: Correction strength<br>A higher value indicates a lower correction strength                                                                                                                                                                                                                                                                                            | DistortionRatio=-205                                                                                                                                                          |                                                                                                                                 |
| bAspect: Whether to maintain the aspect ratio<br>1: Maintain the aspect ratio<br>0: Do not maintain the aspect ratio and retain the maximum field of view                                                                                                                                                                                                                     | bAspect=0<br>DistortionRatio=-165                                                                                                                                             |                                                                                                                                |
| bAspect=0, XRatio, YRatio<br>XRatio: Horizontal field of view reservation magnitude<br>YRatio: Vertical field of view reservation magnitude<br>bAspect=1: Enable XYRatio<br>XYRatio: The field of view reservation magnitude in scene where the aspect ratio is maintained<br>Note: 100 is the maximum field of view retained, 0 is 2/3 of the maximum field of view retained | bAspect=0, XRatio=20<br>bAspect=1, XRatio=20                                                                                                                                  |                                                                                                                                |
|                                                                                                                                                                                                                                                                                                                                                                               | 8                                                                                                                                                                             |                                                                                                                                                                                                                    |

### 3.2.2.2 Example of Correction of Pincushion Distortion

Table 3.4: Instructions for correction of pincushion distortion

| Parameter Description                                                                                                                    | Parameter Configuration                                                                                                                                                      | Image Examples                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Typical Configuration<br>Distortion center overlaps with image center<br>Maintain the aspect ratio<br>Maintain the maximum field of view | Width=1920<br>Height=1080<br>OutWidth=1920<br>OutHeight=1080<br>CenterXOffset/CenterYOffset=0<br>DistortionRatio=500<br>bAspect=1<br>XYRatio=100<br>XRatio=100<br>YRatio=100 | Before correction<br><br>After correction<br> |

## 3.2.3 Free Angle Rotation

Table 3.5: Configuration of Free Angle Rotation

| Configuration Parameter | Configuration Range | Parameter Meaning                                                        |
|-------------------------|---------------------|--------------------------------------------------------------------------|
| CenterXOffset           | -511~+511           | Horizontal offset of the center of rotation from the center of the image |
| CenterYOffset           | -511~+511           | Vertical offset of the center of rotation from the center of the image   |
| OutWidth                | 480~8192            | Width of the output image                                                |
| OutHeight               | 360~8192            | Height of the output image                                               |

## 3.2.4 Data Flow Chart

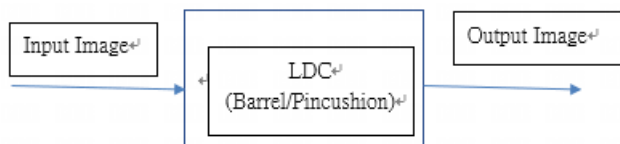


Fig. 3.5: LDC (Lens Distortion Correction flowchart)

# 4 Calibration Tool

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## 4.1 Instruction

Please refer to the PQ tool to make online adjustments to obtain the best model parameters.